OLIST SELLER AND CUSTOMER ANALYSIS

(BRAZILIAN E-COMMERCE PUBLIC DATASET BY OLIST)

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INTRODUCTION



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Education

Universitas Gadjah Mada
Electrical Engineering
Graduated at Mei 2025

Work Experience

Project Overhaul PLTU Sebalang
Instrument & Control Technician
June-August 2025

PROJECT OVERVIEW

- Implementation of the Web-Based Internet of Things in Building Management Systems Aspects of Fire Protection and HVAC

  Tools : Arduino, Thingspeak
- Employee Performance Evaluation: Dashboard performa pegawai dan recomendasi untuk meningkatkan performa

  Tools : MS. Excel
- Pre Processing Project: Project cleaning data dan mendeteksi masalah selama proses cleaning

  Tools : Jupyter Notebook (Python)
- Exploratory Data Analysis: melihat pola statistic pada suatu data

  Tools : Jupyter Notebook (Python)

MAIN PROJECT

- Problem Statement
  - Data Understanding
    - Data Preparation
      - Visualization
        - Recommendation

PROBLEM STATEMENT

- Quick Response Code Indonesian Standard (QRIS)

Kasir memasukkan nominal transaksi pada sistem POS, kemudian merchant menampilkan kode QR yang dapat dipindai oleh pelanggan melalui aplikasi mobile banking atau e-wallet. Setelah pelanggan memindai QR dan mengonfirmasi pembayaran, aplikasi pembayaran akan memproses transaksi melalui jaringan QRIS untuk mengecek saldo, mengotorisasi pembayaran, dan meneruskan dana melalui pihak bank atau penyedia pembayaran yang terhubung.

Pada project kali ini, akan mencoba menjadi ‘jaringan QRIS’ yang akan mengabungkan beberapa sumber data. Hasilnya berupa gabungan table yang dapat dianalisa lebih lanjut.

DATA UNDERSTANDING

- Data Set : Brazilian E-Commerce Public Dataset by Olist
- Table Using :

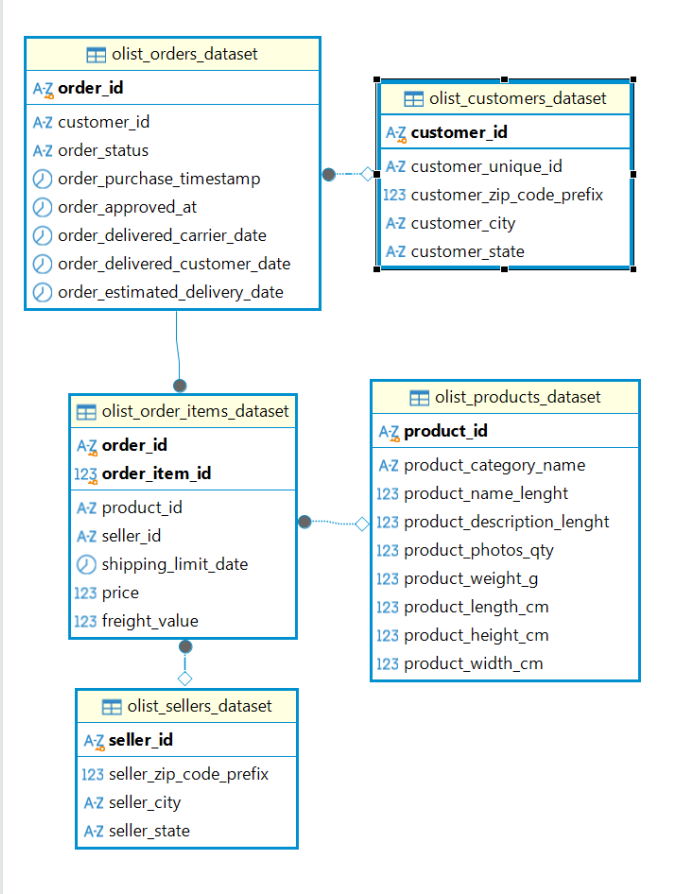
  olist_orders_dataset.csv

  olist_order_items_dataset.csv

  olist_products_dataset.csv

  olist_sellers_dataset.csv

  olist_customers_dataset.csv



OLIST_ORDERS_DATASET.CSV

- Row : 99441
- Missing value

order_id_null	customer_id_null	order_status_null	purchase_null	approved_null	carrier_null	delivered_null	estimated_null
0	0	0	0	160	1,783	2,965	0

- Data type

Column Name	#	Data type
order_id	1	varchar(50)
customer_id	2	varchar(50)
order_status	3	varchar(50)
order_purchase_timestamp	4	varchar(50)
order_approved_at	5	varchar(50)
order_delivered_carrier_date	6	varchar(50)
order_delivered_customer_date	7	varchar(50)
order_estimated_delivery_date	8	varchar(50)

- Column description

Kolom	Deskripsi
order_id	ID unik untuk setiap pesanan
customer_id	ID customer yang melakukan pesanan
order_status	Status pesanan, misalnya delivered, shipped, canceled
order_purchase_timestamp	Waktu customer melakukan pemesanan
order_approved_at	Waktu pesanan/pembayaran disetujui
order_delivered_carrier_date	Waktu pesanan diserahkan ke pihak kurir
order_delivered_customer_date	Waktu pesanan diterima customer
order_estimated_delivery_date	Estimasi tanggal pesanan sampai ke customer

OLIST_ORDER_ITEMS_DATASET.CSV

- Row : 112650
- Missing value

123 order_id_null	123 order_item_id_null	123 product_id_null	123 seller_id_null	123 shipping_limit_null	123 price_null	123 freight_null
0	0	0	0	0	0	0

- Data type

Column Name	#	Data type
A-Z order_id	1	varchar(50)
123 order_item_id	2	int4
A-Z product_id	3	varchar(50)
A-Z seller_id	4	varchar(50)
A-Z shipping_limit_date	5	varchar(50)
123 price	6	float4
123 freight_value	7	float4

- Column description

Kolom	Deskripsi
order_id	ID unik pesanan yang menghubungkan item dengan tabel orders
order_item_id	Nomor urut item dalam satu order
product_id	ID produk yang dibeli
seller_id	ID seller yang menjual produk
shipping_limit_date	Batas waktu seller untuk menyerahkan barang ke pihak logistik
price	Harga produk pada item tersebut
freight_value	Biaya pengiriman untuk item tersebut

OLIST_PRODUCTS_DATASET.CSV

- Row : 32951
- Missing value

product_id_null	category_null	name_length_null	description_length_null	photos_null	weight_null	length_null	height_null	width_null
0	0	610	610	610	2	2	2	2

- Data type

Column Name	#	Data type
product_id	1	varchar(50)
product_category_name	2	varchar(50)
product_name_lenght	3	int4
product_description_lenght	4	int4
product_photos_qty	5	int4
product_weight_g	6	int4
product_length_cm	7	int4
product_height_cm	8	int4
product_width_cm	9	int4

- Column description

Kolom	Deskripsi
product_id	ID unik produk
product_category_name	Kategori produk dalam bahasa Portugis
product_name_lenght	Panjang nama produk dalam jumlah karakter
product_description_lenght	Panjang deskripsi produk dalam jumlah karakter
product_photos_qty	Jumlah foto produk
product_weight_g	Berat produk dalam gram
product_length_cm	Panjang produk dalam cm
product_height_cm	Tinggi produk dalam cm
product_width_cm	Lebar produk dalam cm

OLIST_SELLERS_DATASET.CSV

- Row : 3095
- Missing value

123 seller_id_null	123 zip_null	123 city_null	123 state_null
0	0	0	0

- Data type

Column Name	#	Data type
A-Z seller_id	1	varchar(50)
123 seller_zip_code_prefix	2	int4
A-Z seller_city	3	varchar(50)
A-Z seller_state	4	varchar(50)

- Column description

Kolom	Deskripsi
seller_id	ID unik seller
seller_zip_code_prefix	Kode pos seller
seller_city	Kota tempat seller berada
seller_state	Kode negara bagian seller

OLIST_CUSTOMERS_DATASET.CSV

- Row : 99441
- Missing value

123 customer_id_null	123 unique_id_null	123 zip_null	123 city_null	123 state_null
0	0	0	0	0

- Data type

Column Name	#	Data type
A-Z customer_id	1	varchar(50)
A-Z customer_unique_id	2	varchar(50)
123 customer_zip_code_prefix	3	int4
A-Z customer_city	4	varchar(50)
A-Z customer_state	5	varchar(50)

- Column description

Kolom	Deskripsi
customer_id	ID customer yang terhubung ke tabel orders
customer_unique_id	ID unik untuk mengidentifikasi customer yang sama di beberapa order
customer_zip_code_prefix	Kode pos area customer
customer_city	Kota tempat customer berada
customer_state	Kode negara bagian customer

DATA PREPARATION

- Data types
- Duplicates
- Primary and Foreign key
- Missing Value
- Outlier

DATA TYPES

Query (olist_orders_dataset)

```
--Memperbaiki tipe data
alter table olist_orders_dataset
alter column order_purchase_timestamp
type timestamp
using order_purchase_timestamp::timestamp

alter table olist_orders_dataset
alter column order_approved_at
type timestamp
using nullif(trim(order_approved_at), '')::timestamp

alter table olist_orders_dataset
alter column order_delivered_carrier_date
type timestamp
using nullif(trim(order_delivered_carrier_date), '')::timestamp

alter table olist_orders_dataset
alter column order_delivered_customer_date
type timestamp
using nullif(trim(order_delivered_customer_date), '')::timestamp

alter table olist_orders_dataset
alter column order_estimated_delivery_date
type date
using order_estimated_delivery_date::date
```

Before

Column Name	#	Data type
order_id	1	varchar(50)
customer_id	2	varchar(50)
order_status	3	varchar(50)
order_purchase_timestamp	4	varchar(50)
order_approved_at	5	varchar(50)
order_delivered_carrier_date	6	varchar(50)
order_delivered_customer_date	7	varchar(50)
order_estimated_delivery_date	8	varchar(50)

After

Column Name	#	Data type
order_id	1	varchar(50)
customer_id	2	varchar(50)
order_status	3	varchar(50)
order_purchase_timestamp	4	timestamp
order_approved_at	5	timestamp
order_delivered_carrier_date	6	timestamp
order_delivered_customer_date	7	timestamp
order_estimated_delivery_date	8	date

DATA TYPES

Query (olist_orders_items_dataset)

```
alter table olist_order_items_dataset
alter column shipping_limit_date
type timestamp
using shipping_limit_date::timestamp
```

Before

Column Name	#	Data type
order_id	1	varchar(50)
order_item_id	2	int4
product_id	3	varchar(50)
seller_id	4	varchar(50)
shipping_limit_date	5	varchar(50)
price	6	float4
freight_value	7	float4

After

Column Name	#	Data type
order_id	1	varchar(50)
order_item_id	2	int4
product_id	3	varchar(50)
seller_id	4	varchar(50)
shipping_limit_date	5	timestamp
price	6	float4
freight_value	7	float4

DUPLICATES

```sql
--Cek Duplikat
select order_id, product_id, count(*) AS jumlah
from olist_order_items_dataset
group by order_id, order_item_id
having count(*) > 1

select order_id, count(*) as jumlah
from olist_orders_dataset
group by order_id
having count(*) > 1

select customer_id, count(*) as jumlah
from olist_customers_dataset
group by customer_id
having count(*) > 1

select product_id, count(*) as jumlah
from olist_products_dataset
group by product_id
having count(*) > 1

select seller_id, count(*) as jumlah
from olist_sellers_dataset
group by seller_id
having count(*) > 1
```

Pengecekan duplikat dilakukan untuk menentukan primary dan foreign key suatu tabel

PRIMARY AND FOREIGN KEY

```sql
--Primary dan Foreign key
alter table olist_order_items_dataset
add primary key (order_id, order_item_id)

alter table olist_order_items_dataset
add foreign key (order_id)
references olist_orders_dataset(order_id),
add foreign key (product_id)
references olist_products_dataset(product_id),
add foreign key (seller_id)
references olist_sellers_dataset(seller_id)
```

```sql
alter table olist_orders_dataset
add primary key (order_id),
add foreign key (customer_id)
references olist_customers_dataset (customer_id)
```

```sql
alter table olist_customers_dataset
add primary key (customer_id)

alter table olist_products_dataset
add primary key (product_id)

alter table olist_sellers_dataset
add primary key (seller_id)
```

MISSING VALUE

(Salah satu contoh query)

```
select
count(*) filter (where seller_id is null) as seller_id_null,
count(*) filter (where seller_zip_code_prefix is null) as zip_null,
count(*) filter (where seller_city is null) as city_null,
count(*) filter (where seller_state is null) as state_null
from olist_sellers_dataset
```

order_id_null	customer_id_null	order_status_null	purchase_null	approved_null	carrier_null	delivered_null	estimated_null
0	0	0	0	160	1,783	2,965	0

order_id_null	order_item_id_null	product_id_null	seller_id_null	shipping_limit_null	price_null	freight_null
0	0	0	0	0	0	0

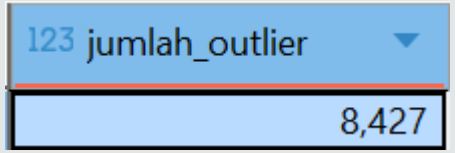
product_id_null	category_null	name_length_null	description_length_null	photos_null	weight_null	length_null	height_null	width_null
0	0	610	610	610	2	2	2	2

seller_id_null	zip_null	city_null	state_null
0	0	0	0

customer_id_null	unique_id_null	zip_null	city_null	state_null
0	0	0	0	0

OUTLIERS (PRICE)

```sql
--Outlier kolom price
--q1 dan q3
with batas as (
select
percentile_cont(0.25) within group (order by price) as q1,
percentile_cont(0.75) within group (order by price) as q3
from olist_order_items_dataset)
-- baris dengan outlier
select o.*
from olist_order_items_dataset o
cross join batas b
where o.price < b.q1 - 1.5 * (b.q3 - b.q1)
or o.price > b.q3 + 1.5 * (b.q3 - b.q1)
order by o.price desc
-- jumlah outlier
select count(*) as jumlah_outlier
from olist_order_items_dataset, batas
where price < q1 - 1.5 * (q3 - q1)
or price > q3 + 1.5 * (q3 - q1)
```



Karna sangat mungkin price bernilai besar, maka untuk nilai outlier “price” tidak dilakukan penanganan (dibiarkan)

OUTLIERS (WEIGHT)

```sql
--Outlier weight
--batas q1 dan q3
with batas as (
select
percentile_cont(0.25) within group (order by product_weight_g) as q1,
percentile_cont(0.75) within group (order by product_weight_g) as q3
from olist_products_dataset)
-- baris dengan outlier
select o.*
from olist_products_dataset o
cross join batas b
where o.product_weight_g < b.q1 - 1.5 * (b.q3 - b.q1)
or o.product_weight_g > b.q3 + 1.5 * (b.q3 - b.q1)
order by o.product_weight_g desc
--jumlah outlier
select count(*) as jumlah_outlier
from olist_products_dataset, batas
where product_weight_g < q1 - 1.5 * (q3 - q1)
or product_weight_g > q3 + 1.5 * (q3 - q1)
```

123 jumlah_outlier
4,551

Karna sangat mungkin berat produk bernilai besar, maka untuk nilai outlier “weight” tidak dilakukan penanganan (dibiarkan)

OUTLIERS (VOLUME)

```sql
--Outlier volume prouct (length*height*width)
--mencari volume
with product_volume as (
select *,
product_length_cm *
product_height_cm *
product_width_cm as volume_cm3
from olist_products_dataset
where product_length_cm is not null
and product_height_cm is not null
and product_width_cm is not null
),
-- batas q1 dan q3
batas as (
select
percentile_cont(0.25) within group (order by volume_cm3) as q1,
percentile_cont(0.75) within group (order by volume_cm3) as q3
from product_volume
)
--baris dengan outlier
--select
--pv.*
--from product_volume pv
--cross join batas b
--where pv.volume_cm3 < b.q1 - 1.5 * (b.q3 - b.q1)
--or pv.volume_cm3 > b.q3 + 1.5 * (b.q3 - b.q1)
--order by pv.volume_cm3 desc
--jumlah outlier
select count(*) as jumlah_outlier
from product_volume, batas
where volume_cm3 < q1 - 1.5 * (q3 - q1)
or volume_cm3 > q3 + 1.5 * (q3 - q1)
```

123 jumlah_outlier
3,262

Karna sangat mungkin volume bernilai besar, maka untuk nilai outlier “volume” tidak dilakukan penanganan (dibiarkan)

VISUALIZATION

order_status	jumlah
delivered	96,478
shipped	1,107
canceled	625
unavailable	609
invoiced	314
processing	301
created	5
approved	2

Order Status	Deskripsi
delivered	Pesanan sudah diterima customer
shipped	Pesanan sedang dikirim
canceled	Pesanan dibatalkan
unavailable	Pesanan tidak dapat dipenuhi karena barang tidak tersedia atau proses tidak dapat dilanjutkan
invoiced	Invoice pesanan sudah dibuat
processing	Pesanan sedang diproses oleh seller
created	Pesanan baru dibuat
approved	Pesanan/pembayaran sudah disetujui

Dari total 99441 order, sebesar 97.02% sudah sampai ke customer(delivered). Ini menunjukan performa yang cukup bagus dalam melakukan jual beli barang secara online

Namun, sebesar 0.62% order dibatalkan (cancel) dan 0.61% order tidak dapat dilanjutkan (unavailable). Walaupun tergolong kecil, Perusahaan tetap perlu memastikan penyebab order tersebut tidak selesai.

PRODUCT

Show SQL | Enter a SQL expression to filter results (use Ctrl+Space)

	product_category_name	jumlah	revenue
1	cama_mesa_banho	11,072	1,034,120
2	beleza_saude	9,583	1,246,250
3	esporte_lazer	8,534	966,407
4	moveis_decoracao	8,245	719,077
5	informatica_acessorios	7,725	896,606
6	utilidades_domesticas	6,873	623,795
7	relogios_presentes	5,930	1,189,030
8	telefonia	4,488	313,214
9	ferramentas_jardim	4,307	475,540
10	automotivo	4,183	584,157
11	brinquedos	4,059	473,957
12	cool_stuff	3,764	618,315
13	perfumaria	3,378	393,526
14	bebes	3,016	405,548
15	eletronicos	2,749	156,756

Produk *cama mesa baho* merupakan produk yang paling banyak terjual, tetapi produk *beleza saude* menjadi produk penghasil revenue terbesar

SELLER

	seller_id	total_order	canceled_order	cancellation_rate	estimated_price_lost
1	cc419e0650a3c5ba77189a1882b7556a	1,706	9	0.53	104,289
2	6560211a19b47992c3666cc44a7e94c0	1,854	7	0.38	123,305
3	620c87c171fb2a6dd6e8bb4dec959fc6	740	6	0.82	114,774
4	c3867b4666c7d76867627c2f7fb22e21	245	5	2.13	37,153.4
5	81783131d2a97c8d44d406a4be81b5d9	13	5	38.46	1,782.54
6	a416b6a846a11724393025641d4edd5e	162	5	3.09	25,670.7
7	8444e55c1f13cd5c179851e5ca5ebd00	99	4	4.12	22,873.2
8	ffff564a4f9085cd26170f4732393726	20	4	33.33	1,426.3
9	b2ba3715d723d245138f291a6fe42594	337	4	1.2	51,676.8
10	4c8b8048e33af2bf94f2eb547746a916	23	4	17.39	2,020.73
11	7e93a43ef30c4f03f38b393420bc753a	336	4	1.22	176,432
12	1127b7f2594683f2510f1c2c834a486b	114	4	3.51	6,851.33
13	3d871de0142ce09b7081e2b9d1733cb1	1,080	4	0.37	94,914
14	dbc22125167c298ef99da25668e1011f	406	4	0.99	33,776.9
15	2138ccb85b11a4ec1e37afbd1c8eda1f	396	3	0.76	12,357.1
16	6973a06f484aacf400ece213dbf3d946	95	3	3.16	11,011.1
17	7040e82f899a04d1b434b795a43b4617	212	3	1.42	9,738.2
18	0be8ff43f22e456b4e0371b2245e4d01	161	3	1.86	11,805.5
19	bacb1f0ed56ad24198f5810d2b3fe9a5	42	3	7.14	6,940.45
20	23d7c96d4a1160db1c726b248601b25a	53	3	5.66	11,766.9
21	46dc3b2cc0980fb8ec44634e21d2718e	521	3	0.59	128,112
22	f5a590cf36251cf1162ea35bef76fe84	117	3	2.56	53,201.8
23	0adac9fbd9a2b63cccaac4f8756c1ca8	110	3	2.73	15,961
24	2e3be8a987a30d7544dbbda6861cc14e	18	3	17.65	4,430.84
25	c4f7fee5b0db50e87766f5a4d1b1b758	8	3	37.5	11,961.8
26	f457c46070d02cadd8a68551231220dd	199	3	1.51	19,093
27	a1043bafd471dff536d0c462352beb48	718	3	0.42	101,901
28	2a84855fd20af891be03bc5924d2b453	164	3	1.83	15,819.1
29	90d4125885ab6c86e8820a722be71974	3	3	100	1,299.55
30	a673821011d0cec28146ea42f5ab767f	129	2	1.57	13,863.7

Jumlah order cancel tertinggi, tidak menunjukan performa yang buruk jika dilihat dari cancellation rate dan juga tidak memberikan price lost yang besar. Perusahaan bisa lebih focus menangani seller dengan cancellation rate dan price lost yang lebih besar.

CUSTOMER

	customer_unique_id	jumlah_transaksi	total_barang	max_barang_satu_transaksi	total_spend	total_freight	freight_persen
1	8d50f5eadf50201ccdcedfb9e2ac8455	16	16	1	729.62	172.42	23.6314788461
2	3e43e6105506432c953e165fb2acf44c	9	14	4	1,000.85	171.82	17.1674072742
3	1b6c7548a2a1f9037c1fd3ddfed95f33	7	9	2	809.21	149.8	18.5118839145
4	ca77025e7201e3b30c44b472ff346268	7	12	4	806.61	316.11	39.1899466515
5	6469f99c1f9dfae7733b25662e7f1782	7	9	2	664.2	94.63	14.247213304
6	47c1a3033b8b77b3ab6e109eb4d5fdf3	6	7	2	836.6	107.61	12.86277771
7	dc813062e0fc23409cd255f7f53c7074	6	11	4	687.76	345.86	50.2878904343
8	63cfc61cee11cbe306bff5857d00bfe4	6	11	3	579.81	246.51	42.5156533718
9	f0e310a6839dce9de1638e0fe5ab282a	6	6	1	438.09	102.6	23.4198436141
10	12f5d6e1cbf93dafd9dcc19095df0b3d	6	6	1	58.4	52.32	89.5890474319
11	fe81bb32c243a86b2f86fbf053fe6140	5	5	1	1,535.4	55.36	3.6055751145
12	5e8f38a9a1c023f3db718edcf926a2db	5	5	1	994.8	119.38	12.0004013181
13	56c8638e7c058b98aae6d74d2dd6ea23	5	7	2	633.97	172.5	27.2094905376
14	394ac4de8f3acb14253c177f0e15bc58	5	6	2	633.16	112.25	17.7285373211
15	de34b16117594161a6a89c50b289d35a	5	5	1	486	44.38	9.1316871345
16	4e65032f1f574189fb793bac5a867bbc	5	6	2	404.57	85.84	21.2175905704
17	35ecdf6858edc6427223b64804cf028e	5	7	2	393.86	210.04	53.3285915852
18	74cb1ad7e6d5674325c1f99b5ea30d82	5	5	1	334.5	116.67	34.878924489
19	b4e4f24de1e8725b74e4a1f4975116ed	5	5	1	200.28	74.67	37.2828006744
20	c8460e4251689ba205045f3ea17884a1	4	24	6	4,080	575.88	14.1147062182
21	4facc2e6fbc2bffab2fea92d2b4aa7e4	4	4	1	1,686.9	73.85	4.3778531253
22	397b44d5bb99eabf54ea9c2b41ebb905	4	6	3	1,650	106.53	6.4563632011
23	f34cd7fd85a1f8baff886edf09567be3	4	4	1	1,047	189.76	18.1241631508
24	08e5b38d7948d37fbb2a59fc5e175ab1	4	4	1	866.79	189.54	21.866889298
25	12d8b5ed661190a3a08183644dfc504d	4	6	2	697.84	160.11	22.9436561465

Customer dengan jumlah transaksi terbanyak (16) tidak menjadi customer dengan spend (729.62) terbesar. Sedangkan customer dengan spend terbesar (4080) tidak sering melakukan transaksi (4). Sehingga belum ditemukan pola khusus untuk mencari customer loyal

CONCLUSION

- Perusahaan memiliki sistem yang cukup bagus dengan 97.02% order sampai ke customer
- Produk dengan tingkat penjualan tertinggi belum tentu menjadi produk dengan penghasil revenue terbesar
- Seller dengan performa terburuk dapat dilihat dari cancellation ratenya
- Belum terlihat pola customer loyal yang konsisten dari data transaksi yang dianalisis.

RECOMENDATION

- Mempertahankan performa order yang sampai ke customer
- Perusahaan dapat memberi peringatan pada seller dengan cancellation rate yang tinggi (diatas 50%)
- Prioritaskan kategori produk dengan melakukan kombinasi penjualan dan revenue tinggi.

TERIMA KASIH

(Terutama buat Maulana Akbar)